and 30° tilted telephone, in the case of tilted handset we observed a lower gain for almost three quadrants.

## IV. Conclusions

This paper investigates in detail the far field of actual cellular telephones held at a realistic position in proximity of the human head—numerically and experimentally. The telephones have been placed at a typical angle of 30° with respect to the vertical. The FDTD method and the measurement system provided results in good agreement, both in shape and gain of the radiation patterns, for all the considered mobile telephones equipped with various different antennas. We have observed a slight decrease of the cellular telephone performance with respect to the vertical position, while a sensible decrease in gain has been found when the phone is held against the human head. This investigation confirms that the FDTD method is a valuable technique to design cellular telephone antennas, being capable of providing accurate results even in challenging conditions involving radiation patterns at tilted angles and modeling of realistic cellular telephones. The technique could be extremely important in the design of mobile telephone since it allows to predict correctly the behavior of the telephone in normal use conditions.

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# Loop-Shield-Loop Shielding Effectiveness

R. C. Hansen and J. Ronald Moser

***Abstract*—The 1967 formulation by Moser of the shielding effectiveness of a metal sheet on a small loop is extended to loop-to-loop coupling. The integration of flux through the receiving loop simply adds another $J_0(ka)$ Bessel function to the Moser integrands; the result is a monotonic variation of coupling with loop spacing, unlike the original results.**

***Index Terms*— Antennas, electromagnetic coupling, loops, near fields, shielding effectiveness.**

## I. Introduction

Electromagnetic shielding of rooms or equipment through use of metal walls or sheets has been important for many decades and is now even more necessary with the burgeoning number of RF transmitters of all types. For plane wave sources, there are well-known simple formulas for transmission through a metal sheet or through a multiple layer wall, using A-B-C-D or equivalent chain matrices. Often, the frequencies of interest are sufficiently low that the metal wall is in the near field of a source or a test antenna. For these cases, the shielding effectiveness (SE) may be significantly different than for the plane wave case.

Moser [1], in a classic paper obtained the SE of a metal wall when the source antenna is an electrically small loop and when the wall is in its near field. The loop magnetic field was expanded into an integral of plane waves; each plane wave component could then be propagated through the wall. The result is a quotient of two integrals involving the $J_1$ Bessel function. Computations were performed by Ryan [2] for two thicknesses of copper, aliminum, and steel walls. Suffice it to say here that excellent agreement existed with measurements; these will be discussed later. Approximate formulas for thin and for thick walls (in skin depths) were given by Bannister [3], [4].

Although Moser formulated the exact near-field produced by the loop and shielding metal wall, the shielding effectiveness formula gave field on the transmitting loop axis. When calculations (described later) are made versus loop-observation point spacing, the shielding shows a "bathtub" shape, that is a minimum shielding occurs for spacing comparable to loop radius. Of most practical interest, however, is coupling between two identical loops; this requires an integration of flux through the receiving loop. In this paper, the modified Moser shielding effectiveness formula is given and results are compared with those from the earlier formula.

Manuscript received June 11, 1996; revised February 9, 1999.

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Publisher Item Identifier S 0018-9375(99)04089-2.

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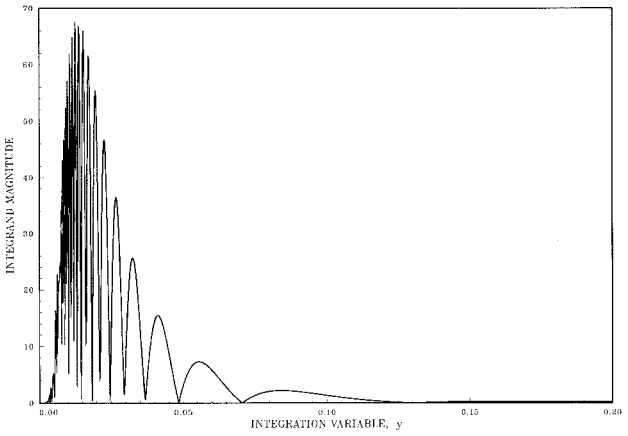



Fig. 1. SE integrand. Radius: 0.5 m; separation: 0.05 m, 1/16 copper, and 1 KHz.

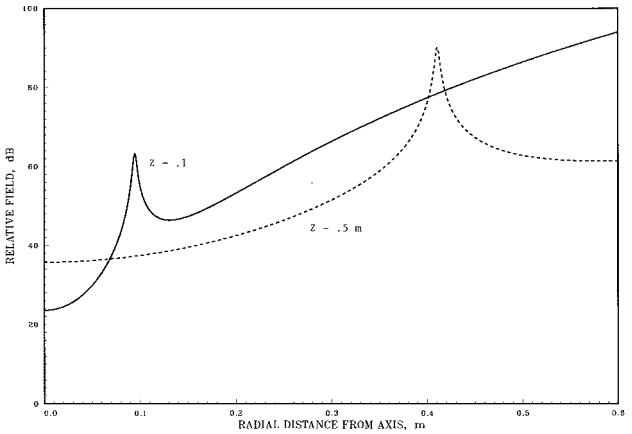



Fig. 2. Moser field variation. Radius $= 0.05$ m, 1/16 copper, 1 KHz.

## II. The Modified Moser SE

To enhance the physical interpretation of the Moser integral for SE, the integration variable was converted to distance. His result [1, eq. (53)] gives the SE at point $(z, \rho)$: the flux through a receiving loop is obtained by integrating over the loop area. Note that result [1, eq. (54)] gives SE on the axis at $(z, 0)$. The modified Moser expression is

$$\mathrm{SE} = \frac{\dfrac{1}{4\mu_r}\displaystyle\int_0^\infty\int_0^a \frac{J_1(a/y)J_0(\rho/y)\exp{-\zeta_0 z/y}}{y^3\zeta_0}\,d\rho\,dy}{\displaystyle\int_0^\infty\int_0^a \frac{\zeta J_1(a/y)J_0(\rho/y)\exp[-\zeta_0(z-t)/y-\zeta t/y]}{y^3[(\zeta+\mu_r\zeta_0)^2-(\zeta-\mu_r\zeta_0)^2\exp -2\zeta t/y]}\,d\rho\,dy}. \tag{1}$$

Here the loop radius is $a, k = 2\pi/\lambda_0$, and the wall parameters are: thickness $t$, relative permeability $\mu_r$, and conductivity $\sigma$. Distance between loops is $z$. Wave numbers are

$$\zeta^2 = 1 - j\omega\mu_0\mu_r\sigma y^2; \quad \zeta_0^2 = 1 - k^2 y^2. \tag{2}$$

As usual $J_0$ and $J_1$ are the Bessel functions.

The $J_1$ Bessel factor oscillates with a slow decay. An integration technique that works well with this type of integrand breaks the range of integration up into segments between zeros of $J_1$. An integrator such as Romberg or Gaussian is used then on each segment. The result is an alternating series and techniques exist for the acceleration of these. The Levin [5] method, which has been used with good success by this author on other problems, was applied here. However, due to the needs of the next section, the Levin approach was abandoned.

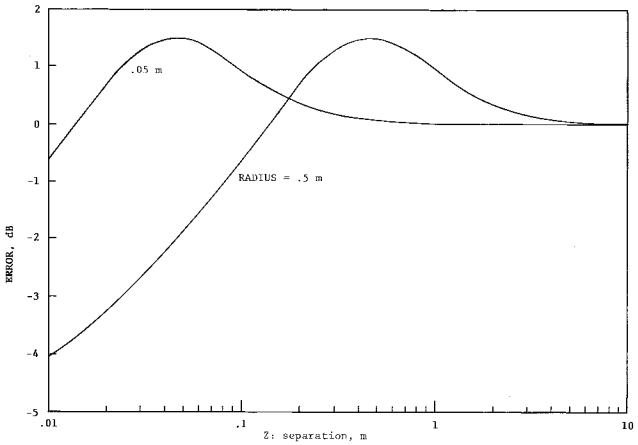



Fig. 3. Error in approximate numerator.

The denominator integrand oscillations for the case of $a = 0.5$ m, $z = 0.05$ m, on-axis field, and frequency of 1 KHz, are shown in Fig. 1. The number of oscillations decreases significantly for larger $z$ and also for small $a$. The numerator integral shows similar behavior. Phase is not shown, but the oscillations in Fig. 1 alternate the sign.

The $y$-integration scheme finally used, divided the integration range into octave segments with a complex GAUSS 128 point integrator used over each segment. A center value was selected based on $z$ and an iterative loop stepped down segments until the relative error in stopping the iteration was less than $10^{-5}$. A similar loop stepped up in segments until error was again less than $10^{-5}$. The final range of variable used is useful, as it indicates the area of the wall where induced currents are significant. For example, a loop–loop spacing of 0.5 m at 1 KHz and the copper wall of Fig. 1 used in integration range of 0.025–25.6 m are valid for loop radii of 0.05–0.5 m. A 64 point GAUSS integrator was used for $\rho$.

To show how the integrand in the denominator behaves, Fig. 2 shows the field versus radial distance for a loop of radius 0.05 m at 1 KHz for spacings of 0.1 and 0.5 m. It can be seen that for spacing comparable to loop diameter, the receiving loop integration is necessary as the field is significantly different than the value on axis.

For on-axis fields at low frequencies, the numerator integral in (1) can be approximated by a simple algebraic form [1]. When the receiving loop integration is introduced, the approximate form may have large errors for spacing comparable to or less than loop radius. Fig. 3 shows the error between the exact numerical integration of the numerator and the approximate result [1, eq. (32)]. Since the denominator integral requires numerical implementation, the numerator integral also should, for most cases of interest.

## III. Results and Comparisons

Integrating the field over the receiving loop removes the anomaly observed earlier, that SE was nonmonotonic. Figs. 4 and 5 compare the Moser SE using on-axis field with the receiving loop-flux (double integration) SE. The loop–loop results show SE approaching 0 dB as spacing approaches zero as expected. Loop-axis results are close as long as the separation between loop and axis point is greater than the loop radius.

## IV. Conclusions

To obtain accurate prediction of loop-to-loop SE double numerical integration is needed for both the numerator and denominator of Moser's expression. And the Bessel function resulting from an

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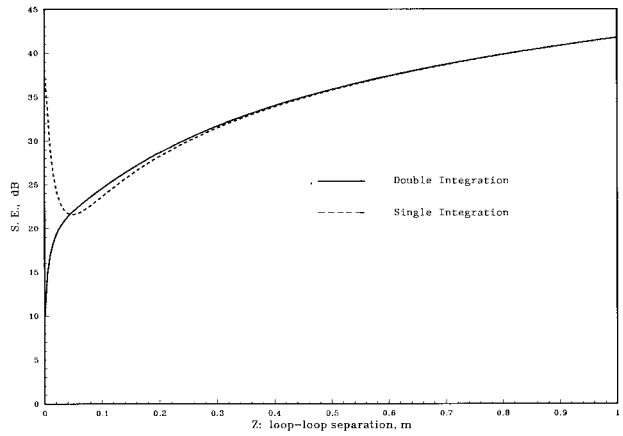



Fig. 4. Loop–loop shielding effectiveness, radius = 0.05 m, 1/16 copper, 1 KHz.

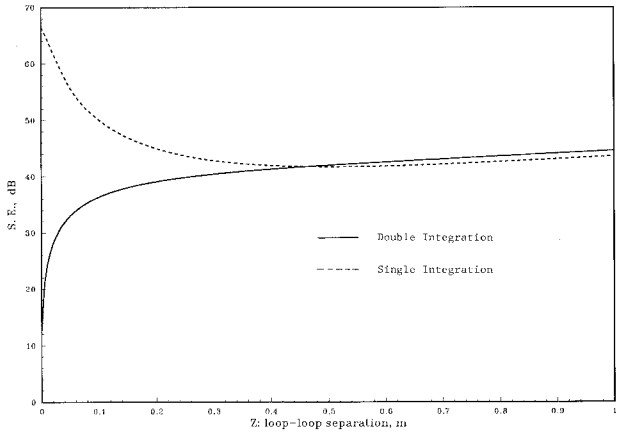



Fig. 5. Loop–loop shielding effectiveness, radius = 0.5 m, 1/16 copper, 1 KHz.

integration over the receiving loop must be included. This provides a monotonic behavior of SE with loop–loop spacing. With modern computers and integrators, numerical results are readily obtained.

## ACKNOWLEDGMENT

The authors would like to thank Dr. T. Griffy for his suggestions.

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# Measuring the Radiated Emissions from a Family of Microprocessors Using a 1-GHz TEM Cell

Kevin P. Slattery, James P. Muccioli, and Terry North

***Abstract*—This paper presents a series of measurements of the radiated emissions from 8 and 16 bit microprocessors. The radiated emissions were measured using a 1-GHz TEM cell that incorporates the device under test (DUT) into the cell structure itself [1]. For the 16-bit processor, samples from each of the manufacturer's identified process corners were measured and compared. Two separate fabrication lines were compared for process variability. Spatial location on the wafer was measured for emissions variation. In addition, emissions were measured for a 16-bit processor as a function of operating temperature. Finally, a comparison was made between discrete implementations of a module digital core and the same circuit implemented as an multichip module (MCM).**

***Index Terms*—Microprocessor, radiated emissions.**

## I. INTRODUCTION

The 1-GHz Fischer TEM cell was developed expressly for the purpose of measuring the emissions from very large scale integration (VLSI) devices in the absence of printed circuit board (PCB) effects. The TEM cell used is described in Society of Automotive Engineers (SAE) specification 1752/3, Integrated Circuit Radiated Emissions Measurement Procedure and is manufactured by Fischer Custom Communications. Essentially, the Fischer TEM cell incorporates the device to be measured, a microprocessor, or any large scale device, into the TEM cell structure itself. By doing so, the device under test has negligible effect upon the cell characteristics [1], [2]. The emissions measured by the TEM cell have been shown to correlate with emissions as measured in a semi-anechoic chamber [1]. The TEM cell is single moded to 1.3 GHz. The devices under test (DUT's) measured and described in this paper were manufactured by Motorola and are typical of devices found in automotive control electronics. Each processor is mounted on a support board that is then placed in the TEM cell (Fig. 1).

All of the DUT support circuitry resides outside the cell; only the DUT is inside. The circuit board itself becomes a part of the cell floor. The PCB is of multilayer construction. The outer layer under the DUT is ground.

## II. MEASUREMENT PROCEDURE AND REPEATABILITY

The repeatability of the Fischer TEM cell was tested at ten different bands in the frequency range. At each point the device was swept 20 consecutive times and the maximum peak was noted. Fig. 2 shows the max–min delta for these points. Repeatability was seen to be

Manuscript received May 1, 1997; revised February 5, 1999.

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